Republic of Tunisia
Ministry of Higher Education and Scientific Research
University of Tunis El Manar
Higher Institute of Computer Science

Design and Development of a Training Management System

Prepared by:

Firas Kahlaoui
Ahmed Chaabane
Seifdine Ben Fradj

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Abstract

This project focuses on the design and development of a web-based training management system intended to improve the organization, monitoring, and evaluation of professional training activities. The existing approach relies on manual processes using paper documents and spreadsheet tools, which leads to inefficiencies, redundancy, and data inconsistency. The proposed system introduces a centralized platform that allows users to manage training sessions, participants, and trainers in a structured and reliable way. It also provides a statistical analysis module to support decision-making. The implementation of this system reduces administrative workload, improves data accuracy, and enhances the overall efficiency of training management.

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Chapter 1

Introduction

1.1 Background and Importance

Professional training is essential for maintaining and improving employee competencies within an organization. It enables individuals to acquire new knowledge, adapt to technological changes, and take on greater responsibilities. For employees, it serves as a vehicle for career advancement and personal growth. For organizations, continuous training leads to improved productivity, better service quality, and stronger employee engagement, ultimately contributing to a competitive advantage in a rapidly evolving market.

1.2 Problem Statement

Despite its importance, many organizations still manage training activities through traditional, manual methods. The reliance on paper-based forms and static Excel spreadsheets introduces significant limitations in scalability and data management. These manual workflows are prone to human error, data duplication, and loss of information, making it extremely difficult to monitor ongoing activities and generate reliable performance indicators.

1.3 Objectives

The objective of this project is to design and implement a centralized web-based application, titled "TrainingMS", that automates the management of training activities. The system aims to:

- Provide a secure platform for data entry and validation.
- Centralize participant, trainer, and session information.
- Simplify the scheduling and planning of training programs.
- Offer analytical tools to visualize and evaluate training outcomes through a statistics dashboard.

By digitizing these processes, the project seeks to improve operational efficiency, ensure data integrity, and support strategic decision-making for organizational stakeholders.

Chapter 2

Project Context and Analysis

2.1 Existing System Analysis

The current system for managing training activities within the organization is largely manual and fragmented. The workflow typically begins with various organizational structures submitting their annual training needs using paper-based documents. These documents include specific details such as the training domain, proposed titles, and lists of potential participants with their professional profiles and departmental affiliations.

Once these needs are collected, the information is manually transcribed into separate Excel workbooks. Each year, a new file is created, which leads to significant challenges in maintaining historical records or tracking the long-term progress of individual employees.

2.2 Identified Problems

The manual process introduces several critical issues that hinder the effectiveness of the training programs:

- **Time Inefficiency**: Repetitive manual data entry and cross-referencing between documents consume substantial administrative resources.
- **Data Inconsistency**: Without automated validation, typographical errors and inconsistent naming conventions lead to duplicate records and unreliable data.
- **Lack of Real-time Tracking**: Monitoring the status of a training session or the attendance of participants requires searching through multiple physical or digital files.
- **Analytical Difficulty**: Generating summary statistics or performance reports is a manual, error-prone task that often yields inaccurate results.

2.3 Proposed Solution: TrainingMS

To address these limitations, we propose the development of "TrainingMS", a centralized web application. This solution focuses on the digital transformation of the training workflow through:

- **Centralization**: A unified database to store all information regarding trainers, participants, formations, and historical data.
- **Automation**: Streamlined processes for session planning, participant enrollment, and automated validation of inputs.
- **Reliability**: Enforcement of data integrity through a structured relational database schema.
- **Decision Support**: A dedicated statistics module that provides real-time visualization of training metrics, enabling managers to evaluate the impact of training investments.

This transition from manual spreadsheets to a robust web platform ensures a scalable, secure, and efficient management environment.

Chapter 3

Design and Modeling

3.1 Actors and Roles

The TrainingMS system defines three distinct user profiles to ensure secure and efficient operations:

- **Simple User**: An operational user responsible for the daily management of training data. Their functionalities include authentication (login/logout), management of formations (creation, editing, deletion, and viewing), participants, and trainers. They also handle assignments, such as enrolling participants and associating trainers with specific sessions.
- **Manager (Responsable)**: An analytical user focused on evaluation and decision-making. Unlike the Simple User, the Manager has **read-only access** to the system's core data and cannot perform any modification, creation, or deletion. Their primary focus is the statistics dashboard, where they analyze formation counts, participant data, and budget distribution through interactive charts and advanced filters.
- **Administrator**: A high-level actor with full control over system configuration. Administrators manage user accounts and core referential data (domains, structures, profiles). By inheritance, the Administrator has access to all functionalities of **both** the Simple User (for data management) and the Manager (for analytical overview).

3.2 Functional Requirements

The application must support the following core capabilities:

- **Authentication**: Secure login and logout using JSON Web Tokens (JWT).
- **CRUD Operations**: Comprehensive management (Create, Read, Update, Delete) for participants, trainers, sessions, and reference tables.
- **Training Planning**: Ability to schedule sessions, assign trainers, and enroll participants.
- **Dashboarding**: Real-time display of aggregated statistics and trends.

3.3 Non-Functional Requirements

To ensure a professional-grade application, several non-functional aspects were prioritized:

- **Security**: Implementation of role-based access control (RBAC) and password encryption.
- **Performance**: Optimization of database queries to handle growing datasets efficiently.
- **Usability**: A responsive and intuitive user interface designed to minimize the learning curve.
- **Data Integrity**: Strict adherence to relational constraints and input validation.

3.4 UML Design

3.4.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the identified actors and the system's functionalities. The primary use cases include "Manage Formations", "Register Participants", and "Consult Dashboard", with administrative tasks such as "Manage Users" reserved for the Admin actor.

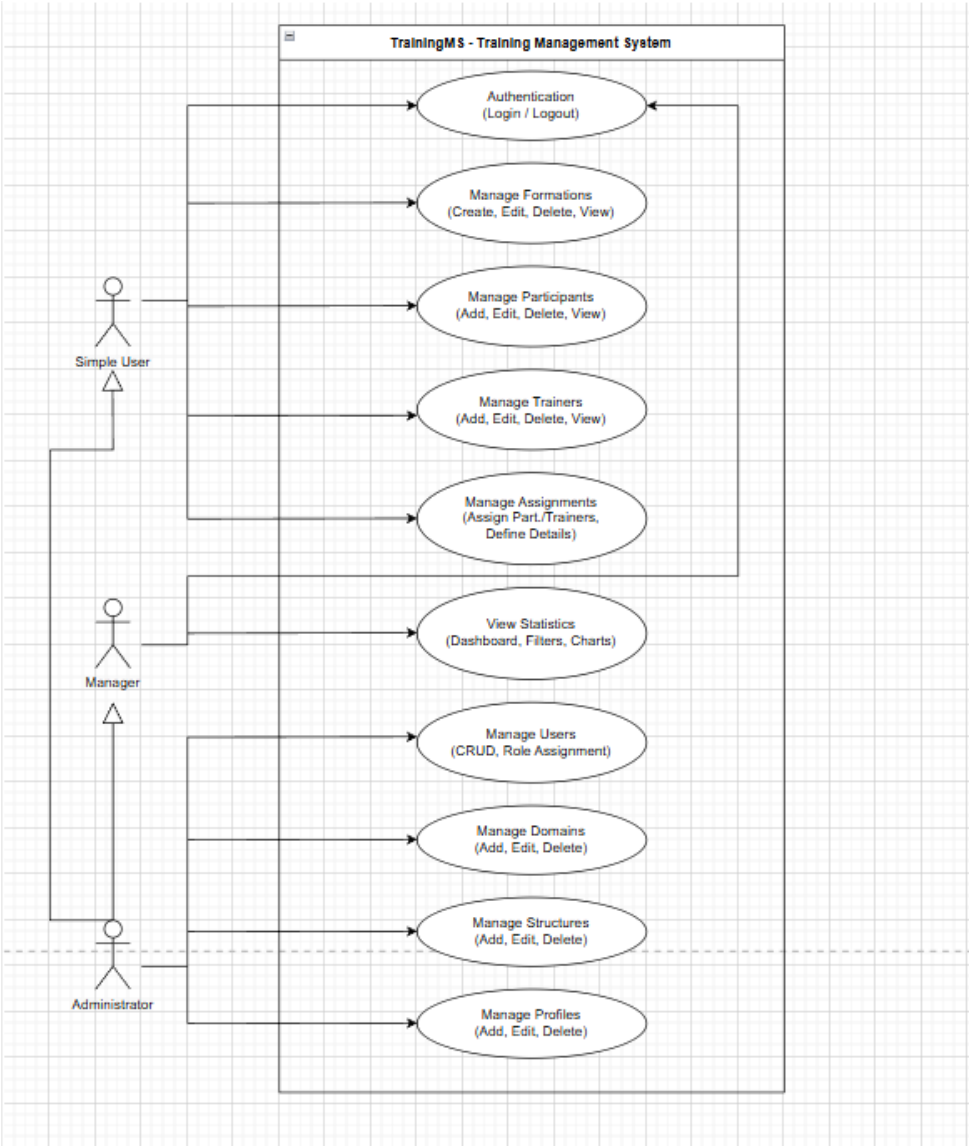

Figure 3.1: TrainingMS Use Case Diagram

3.4.2 Class Diagram

The core logic of the system is modeled through several interrelated entities. The diagram below shows the domain model, including relationships between users, roles, training sessions, participants, and trainers.

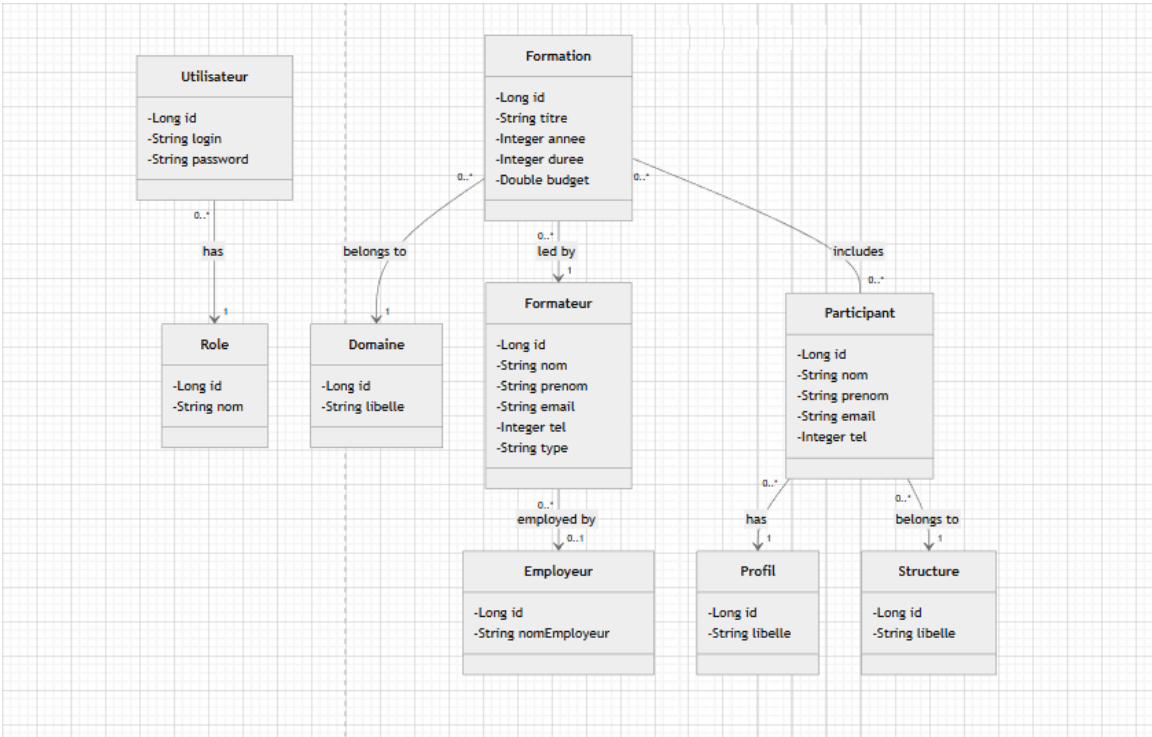

Figure 3.2: TrainingMS System Class Diagram

3.5 Sequence Diagrams

Sequence diagrams illustrate the dynamic behavior of the system and the interactions between objects over time.

3.5.1 User Authentication

This diagram details the exchange between the React frontend, the Security Filter Chain, and the AuthController. It highlights the JWT generation and BCrypt password validation.

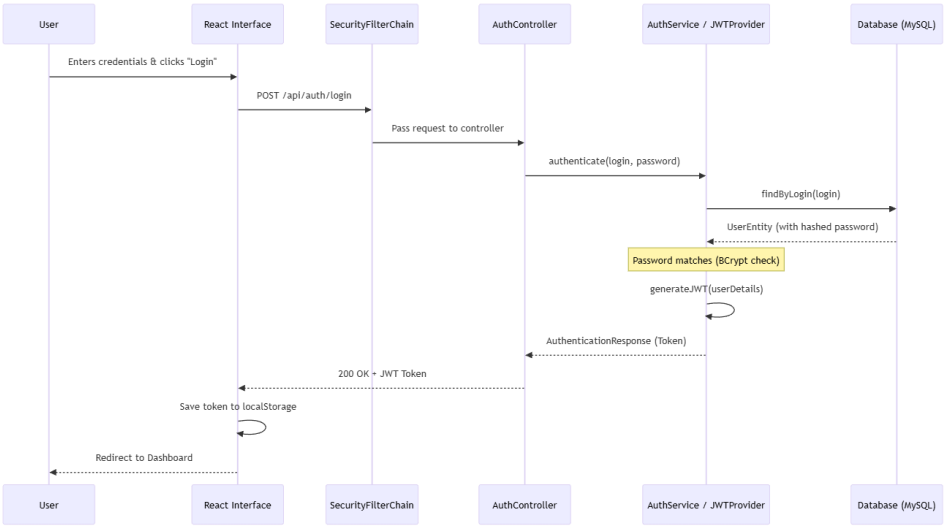

Figure 3.3: Sequence Diagram: User Authentication (Login)

3.5.2 Managing Formations

Illustrates the workflow for creating or updating training sessions, showing the interaction between the FormationController and the Service layer for data persistence.

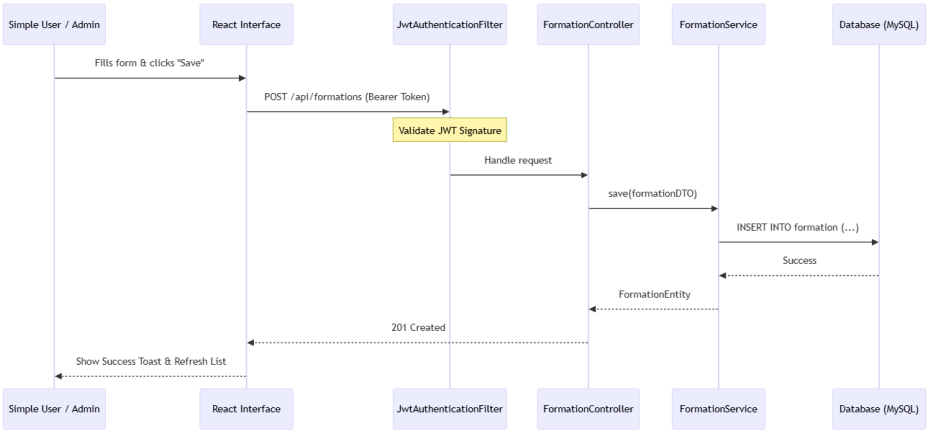

Figure 3.4: Sequence Diagram: Managing Formations (CRUD)

3.5.3 Consulting Dashboard

Describes the analytical flow where the StatsController aggregates data from multiple tables to provide real-time metrics for Managers and Administrators.

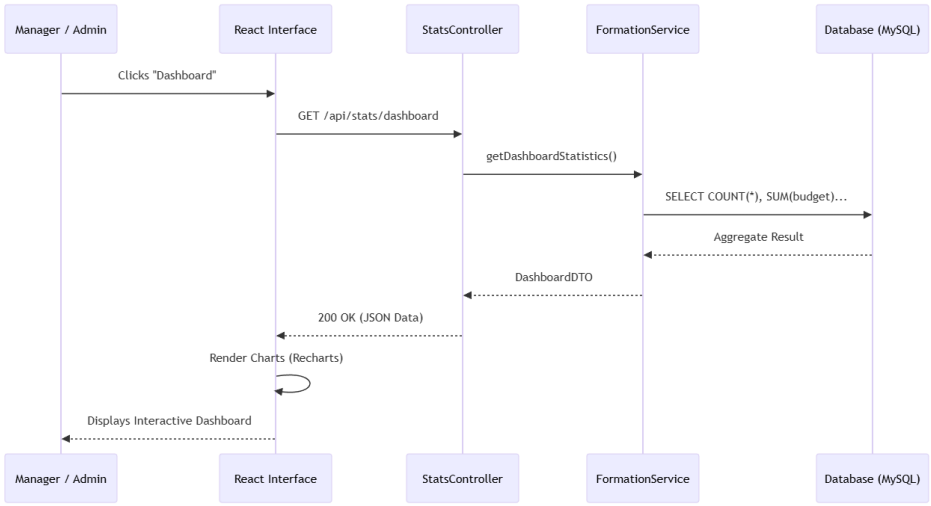

Figure 3.5: Sequence Diagram: Consulting Dashboard (Statistics)

3.5.4 Registering a Participant

Details the process of adding a new participant to the system, including the validation of profile and structure assignments.

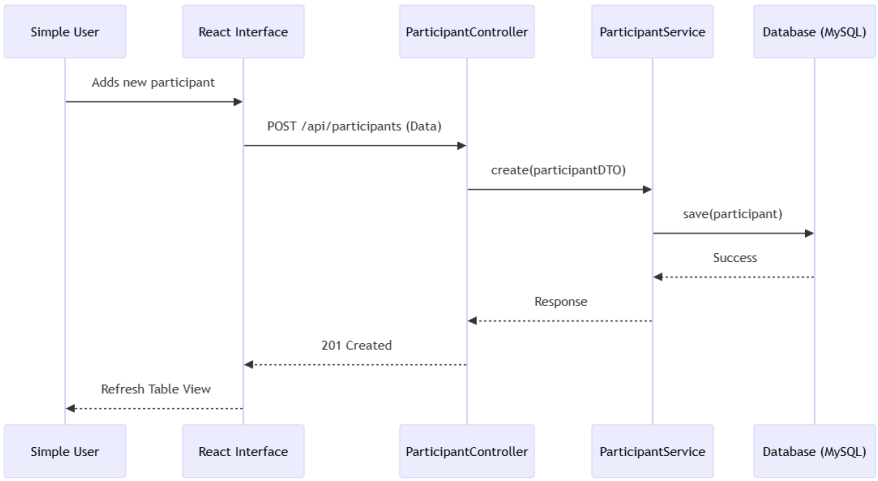

Figure 3.6: Sequence Diagram: Registering a Participant

3.5.5 User Management (Admin Only)

Illustrates the high-level administrative process for managing system accounts, protected by specific role-based security checks.

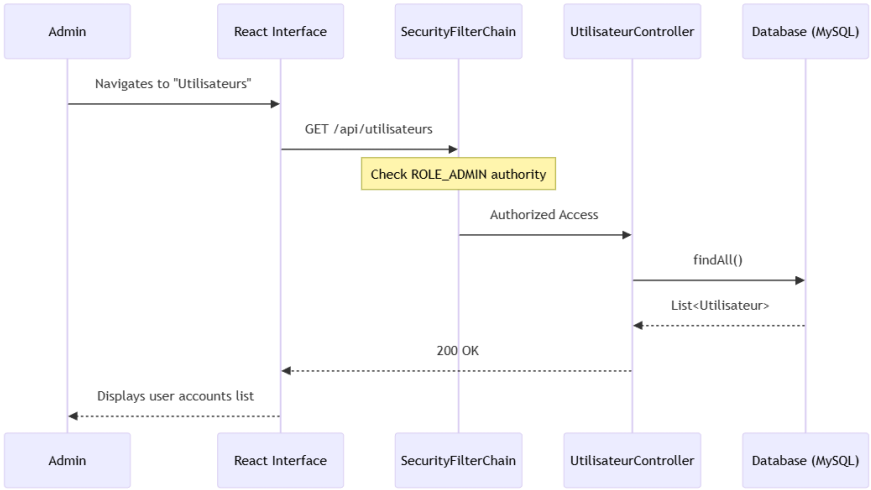

Figure 3.7: Sequence Diagram: User Management (Administration)

3.6 Database Design

The relational schema was designed in MySQL to ensure optimal storage and fast retrieval. The database includes:

- **Tables with Primary Keys**: All tables use auto-incremented integers as unique identifiers.
- **Foreign Key Constraints**: Relationships are enforced through foreign keys (e.g., *idDomain* in the *training session* table).
- **Integrity Rules**: Cascade operations and nullability constraints are defined to prevent orphaned records.

This structured approach guarantees that the data remains consistent throughout the application's lifecycle.

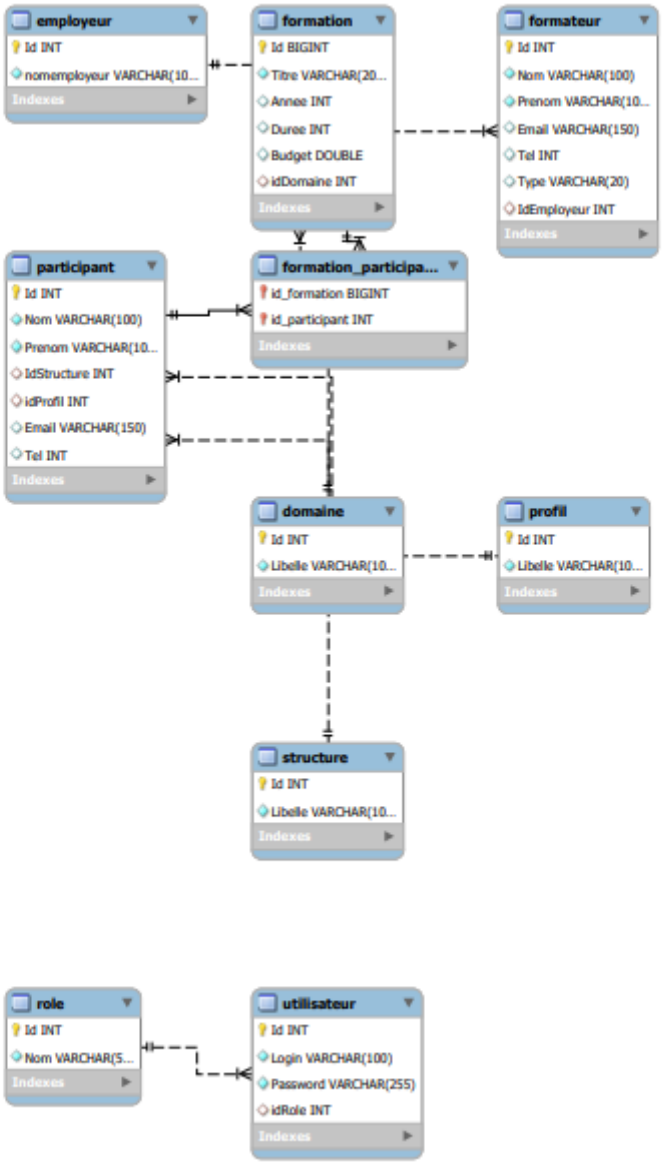

Figure 3.8: TrainingMS Logical Data Model (LDM)

Chapter 4

Implementation and Technologies

4.1 Environment Setup and Technical Requirements

The application development environment was configured with the following specific versions to ensure compatibility and stability:

- **JDK 21**: Utilized for the backend to leverage the latest Java performance improvements and features.
- **Node.js**: Required for building and running the React-based frontend development server.
- **DBMS (MySQL 8.0)**: Used as the relational database engine for data persistence.

4.2 Technology Stack

The application is built using a modern full-stack architecture to ensure scalability and ease of maintenance:

- **Backend (Spring Boot)**: Chosen for its robust security features, dependency injection, and mature ecosystem. It handles the RESTful API and business logic.
- **Frontend (React)**: A component-based library used to build a dynamic and responsive single-page application (SPA).
- **Database (MySQL)**: A reliable relational database management system used to store the structured data of the project.

Figure 4.1: Project Technology Stack

4.3 System Architecture and Security

4.3.1 Security Implementation

Security is a core pillar of TrainingMS. The system implements **Spring Security** to handle the authentication and authorization flow. A custom `SecurityFilterChain` is configured to manage access controls, ensuring that:

- Authentication is stateless using **JSON Web Tokens (JWT)**.
- Password encryption is handled via BCrypt.
- Role-based access control (RBAC) is enforced, restricting sensitive operations to Administrators while allowing read-only access for Managers.

4.3.2 API Documentation

The backend exposes a set of RESTful endpoints. The table below summarizes the key APIs used by the frontend to interact with the system:

Method	**Endpoint**	**Description**
POST	`/api/auth/login`	User authentication and JWT generation.
GET	`/api/formations`	Retrieve the list of all training sessions.
POST	`/api/formations`	Create a new training session (Admin/User).
GET	`/api/stats/dashboard`	Retrieve aggregated data for the dashboard.
GET	`/api/participants`	List all registered participants.

Table 4.1: Core System Endpoints

The backend follows a layered architecture consisting of the Controller layer (REST endpoints), the Service layer (Business logic), and the Repository layer (Data access via Spring Data JPA).

4.4 User Interface Design

The user interface is designed with a focus on productivity. It features a persistent sidebar for navigation, comprehensive data tables with search and pagination capabilities, and modal-based forms for data entry. The responsive layout ensures that the system remains accessible across different device types.

4.5 Statistics Module

A critical component of TrainingMS is the statistics dashboard. It provides interactive charts (Bar, Pie, and Line charts) that visualize data such as:

- Training distribution by domain.
- Evolution of training sessions over time.

- Budget utilization by organizational structure.

The module supports real-time filtering by year, domain, and trainer type. All financial data is formatted in Tunisian Dinars (TND) for local context. The dashboard automatically updates as filters are applied, providing managers with an accurate and instantaneous overview of the training landscape.

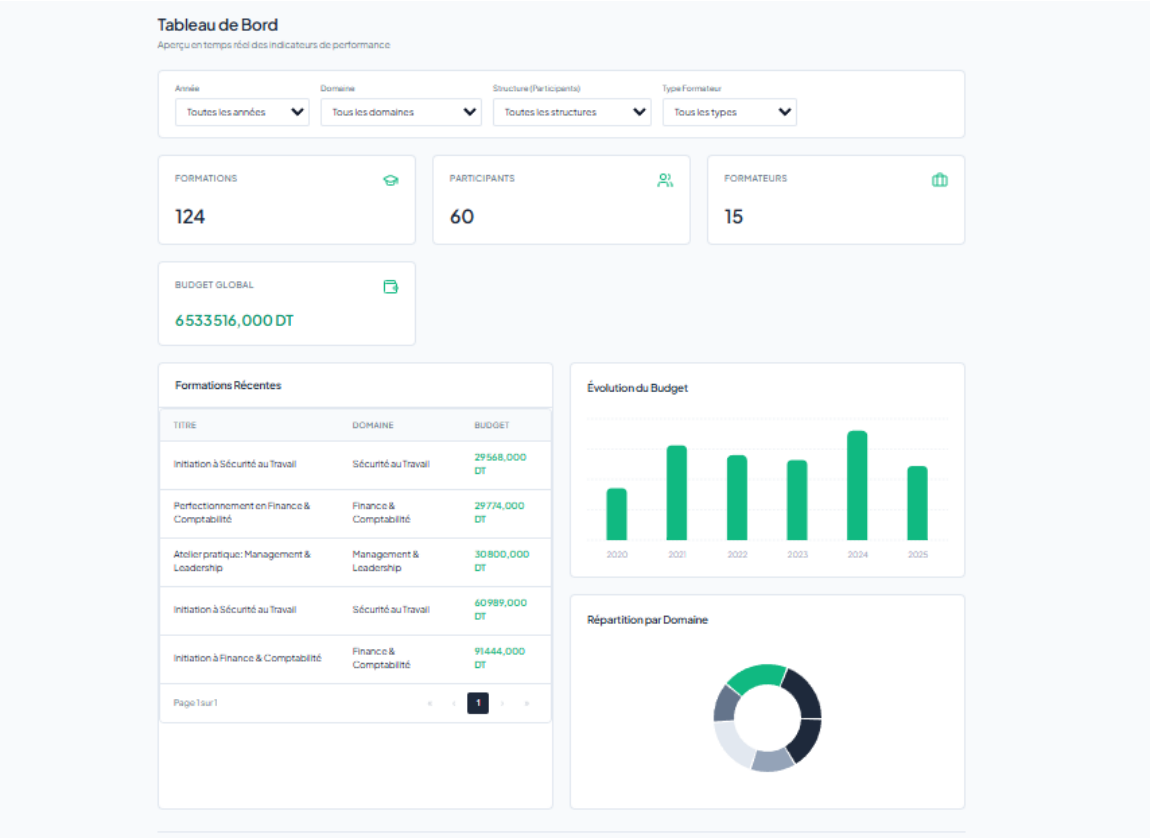

Figure 4.2: TrainingMS Analytical Dashboard

Chapter 5

Testing and Results

5.1 Testing Strategy

To ensure the reliability of the application, a multi-phased testing approach was adopted:

- **Functional Testing**: Each feature (e.g., creating a formation, registering a participant) was tested against its functional requirements to ensure correctness.
- **Validation Checks**: Input fields were tested for edge cases, including empty values, incorrect formats (e.g., invalid emails), and numeric range violations.
- **API Testing**: Backend endpoints were validated using tools like Postman to verify status codes, JSON response structures, and security constraints.

5.2 Results and Discussion

The development of TrainingMS successfully transitioned the organization's training management from a fragmented manual process to a centralized digital platform.

5.2.1 Improvements Over Manual System

The primary achievements include:

- **Data Accuracy**: Centralized storage eliminated the data duplication issues inherent in yearly Excel files.
- **Efficiency**: Search and retrieval times for participant records and session history were reduced from minutes to seconds.
- **Visibility**: The statistics module provided insights into training trends that were previously impossible to generate manually.

5.2.2 Strengths and Limitations

The application's strengths lie in its modular architecture and user-centric design. However, some limitations remain, such as the current lack of a mobile-specific interface and the absence of an automated email notification system for participants. These areas provide opportunities for future enhancements.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project involved the design and development of a "Training Management System" (TrainingMS) to modernize and automate the management of professional training activities. By replacing manual paper and spreadsheet-based processes with a centralized web application, the project addressed critical issues such as data inconsistency, time inefficiency, and the lack of analytical tracking.

The successful implementation of features like role-based access control, CRUD management of entities, and an interactive statistics dashboard demonstrates the value of the platform in improving organizational productivity and data reliability. The project served as a comprehensive exercise in full-stack development, integrating Spring Boot, React, and MySQL into a coherent and functional solution.

6.2 Future Work

While the current version of TrainingMS fulfills the core requirements, several areas for future development have been identified:

- **Advanced Analytics**: Integrating machine learning models to predict future training needs based on employee performance data.
- **Automated Notifications**: Developing a notification engine to send automated reminders and certificates via email or SMS.
- **Mobile Application**: Creating a dedicated mobile version for participants to view their training history and upcoming sessions on the go.
- **Feedback System**: Implementing an evaluation module to collect and analyze participant feedback after each session.

These enhancements will further solidify TrainingMS as a comprehensive tool for workforce development and organizational growth.